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A Review of IoT and Cloud Computing: Issues, Challenges, and Opportunities in Eutrophication Monitoring

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Systematic Review

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Abstract

Eutrophication remains a significant ecological threat to freshwater and coastal environments, often resulting in algal blooms, oxygen depletion, and biodiversity loss. Traditional monitoring methods—such as periodic manual sampling and laboratory analysis—lack the spatial and temporal resolution required for real-time assessment and management. Recent advances in Internet of Things (IoT) and cloud computing offer promising alternatives for continuous, automated water quality monitoring. This review aims to systematically evaluate current research on the integration of IoT and cloud platforms for eutrophication monitoring, focusing on system architectures, communication protocols, storage mechanisms, deployment strategies, and security practices. A total of 5,744 articles were identified through Google Scholar, Web of Science, and Scopus databases. After title and abstract screening and full-text evaluation, 81 articles published between 2015 and 2024 met the inclusion criteria. Descriptive analysis and categorization were performed on various system components, including cloud usage, communication protocols, data storage, deployment platforms, and microcontroller adoption.

Research output on IoT-based eutrophication monitoring has steadily increased, peaking in 2023. The majority of studies originated from Asia-Pacific regions, particularly India (33.33%) and China (13.58%). Most papers were indexed in Google Scholar (57%), with fewer in Scopus (29%) and Web of Science (14%). ThingSpeak (30.86%) and custom platforms (9.88%) were the most used cloud services, while 23.46% of studies did not specify the platform. HTTP (33.33%) and MQTT (12.35%) were the most common communication protocols. In terms of storage, 35.80% employed miscellaneous or unclassified mechanisms, followed by real-time databases (20.99%) and SQL (13.58%). Hybrid cloud-edge deployment was dominant (55.56%), reflecting a need for both scalability and real-time responsiveness. Security practices were inconsistently reported; 49.38% of studies did not specify any mechanism, while encryption (24.69%) and authentication (16.05%) were the most mentioned. The Arduino family (34.57%) and ESP series (23.45%) were the most adopted microcontrollers. Although IoT integration into eutrophication monitoring systems is growing, there are persistent gaps in cloud analytics integration, standardized security implementation, and system documentation. The widespread use of hybrid cloud-edge models suggests an evolving architecture aimed at balancing performance and scalability. However, underutilization of advanced cloud services like predictive analytics (used in only ~ 30% of studies) and inconsistent protocol/reporting standards call for unified frameworks to guide future research and deployment.

1. Introduction

The integration of Internet of Things (IoT) and cloud computing technologies has significantly transformed environmental monitoring by enabling real-time data acquisition, scalable storage, and advanced processing capabilities (Fang et al., 2014; Ajayi et al., 2015). These technologies are increasingly recognized as powerful tools for addressing complex ecological challenges such as eutrophication, where continuous, high-resolution data on nutrient concentrations, dissolved oxygen, and

biological indicators is essential for informed management (Malik & Om, 2017; Monroe & Langford., 2024).

Despite growing interest, the academic literature remains limited in its focused treatment of eutrophication monitoring using combined IoT–cloud frameworks (Ficili et al., Liu et al., 2019; Patil et al., 2012; Matthew et al., 2021). Previous works emphasized the role of wireless sensor networks (WSNs) and embedded systems (ESs) in capturing physical and chemical water quality parameters such as temperature and pH. While these studies demonstrated the value of real-time data for responsive environmental management, they paid limited attention to key eutrophication drivers like nitrates, phosphates, and chlorophyll-a. Moreover, cloud computing was typically relegated to backend storage functions, with minimal discussion on its potential for remote analytics, system interoperability, or predictive modeling (Malik & Om, 2017).

Since 2019, there has been a gradual shift in the literature toward recognizing the role of cloud platforms in managing the scale and complexity of IoT-generated data. Some reviews have explored real-time processing, centralized dashboards, and high-frequency sampling for environmental monitoring. However, applications of these frameworks to freshwater ecosystems—particularly those impacted by eutrophication—remain sparse. The use of cloud computing for ecological modeling, early warnings of harmful algal blooms, and adaptive management based on historical trends has not been widely demonstrated in this context.

Recent literature reflects a growing interest in the integration of Internet of Things (IoT) and cloud computing technologies for environmental and water quality monitoring across diverse contexts. For instance, Kumar and Saini (2022) demonstrated a data-driven IoT-cloud framework for hydropower plant monitoring, emphasizing real-time analytics. Similarly, Garrido-Momparler and Peris (2022) reviewed smart sensor deployment for general environmental monitoring, while Stojanovic and Chaudhary (2023) explored real-time aquaculture monitoring using cloud-based IoT systems. Some studies have begun to address eutrophication directly—Shivaanivarsha et al. (2022) developed a low-cost, multi-parameter lake monitoring system aimed at early eutrophication detection, and Mazibuko et al. (2025) reviewed oxygen dynamics monitoring using IoT sensors in eutrophic waters. Broader overviews, such as that by Nethanani et al. (2025), synthesized challenges and trends in biological water monitoring through IoT-cloud architectures, but without specific focus on nutrient dynamics. Other relevant contributions include the use of smart buoys and satellite data for algal bloom detection (Moron-Lopez et al., 2020), integrated remote sensing strategies (Lai et al., 2024), and AI-driven agricultural water quality monitoring (Mohapatra et al., 2024). However, these studies often focus on single aspects such as hardware, communication, or specific environments (e.g., aquaculture, estuaries, or agriculture) rather than providing a unified analysis tailored to eutrophication monitoring. Unlike these prior works, the present review offers a systematic and focused synthesis of 81 peer-reviewed studies (2015–2024), specifically examining how IoT-cloud systems are technically configured and deployed in the context of freshwater eutrophication. It uniquely combines analysis of microcontroller usage, cloud platform selection,

protocol layering, storage strategies, deployment models, and security mechanisms—areas often fragmented or overlooked in earlier studies.

1.1. Rationale

Eutrophication remains one of the most pressing ecological challenges affecting freshwater systems worldwide. It involves nutrient over-enrichment, leading to harmful algal blooms, oxygen depletion, and long-term degradation of aquatic habitats. The convergence of IoT technologies with cloud computing offers transformative potential for addressing these challenges, enabling real-time sensing, distributed data collection, cloud-based analytics, and scalable system management. However, while IoT-cloud frameworks have been increasingly applied in areas like general water quality monitoring, smart agriculture, and municipal infrastructure, their targeted application to eutrophication monitoring is underrepresented in the literature.

This review is motivated by the need to synthesize the fragmented body of research focused on the integration of IoT and cloud computing in the specific context of eutrophication. Existing studies often overlook nutrient-specific indicators, lack ecological contextualization, and rarely explore security, scalability, or interoperability in depth. Drawing on data from 81 systematically selected articles spanning 2014 to 2024, this review provides a comprehensive evaluation of the technologies, platforms, protocols, and architectures employed. It aims to bridge the gap between technical implementation and environmental application, supporting researchers and practitioners in deploying more effective monitoring systems for eutrophication-prone environments.

1.2. Objectives

The primary objective of this systematic review is to evaluate how IoT and cloud computing technologies are currently being integrated in eutrophication monitoring systems, with an emphasis on their technical composition, performance trade-offs, and future scalability. Specifically, the review aims to:

- Analyze the cloud platforms and communication protocols most frequently employed in reviewed systems, assessing their performance implications and usage trends.
- Evaluate data upload methods, storage mechanisms (e.g., SQL, JSON, real-time databases), and architectural choices (e.g., edge, cloud, hybrid), with a focus on responsiveness, latency, and scalability.
- Examine the extent and consistency of security implementations, including encryption, authentication, and data privacy strategies across reviewed systems.
- Assess microcontroller integration, hardware configurations, and the broader deployment context of IoT systems within freshwater ecosystems.

1.3. Research Questions

To guide the systematic synthesis, this review addresses the following research questions:

- Which cloud platforms, data upload methods, and communication protocols are most commonly used in eutrophication monitoring, and what are their impacts on system performance, efficiency, and interoperability?
- How do data storage mechanisms (e.g., relational, real-time, hybrid), latency, and system architecture (edge/cloud/hybrid) influence the scalability and real-time responsiveness of IoT-based eutrophication monitoring frameworks?
- What types of security mechanisms (e.g., encryption, authentication, access control) are employed across the systems, and how effectively do they safeguard data integrity and privacy?
- What limitations and opportunities exist in current eutrophication monitoring systems regarding microcontroller integration, energy efficiency, and broader deployment across different aquatic environments?

1.4. Research Contribution

This review makes several key contributions to the field of digital environmental monitoring:

- It provides the first systematic review that categorizes IoT–cloud-based eutrophication monitoring systems by platform usage, communication protocol, storage solution, and system architecture (e.g., edge, cloud, hybrid).
- It critically assesses how architectural choices impact responsiveness, latency, and scalability—highlighting that over 55% of systems now favor hybrid deployment models, and nearly 30% of systems fail to specify key performance metrics.
- The review highlights a notable gap in security implementation—49.38% of reviewed studies did not specify any security mechanism—while providing a comparative overview of systems that did employ encryption or authentication.
- It offers an analysis of microcontroller adoption trends, showing dominance of the Arduino family (34.57%) but also underreporting in 28.40% of cases—pointing to a disconnect between sensor hardware and software ecosystems.
- Finally, it outlines critical research gaps and proposes actionable directions for future development, including standardization, privacy enhancement, and real-time analytics for nutrient-specific indicators.

1.5. Research Novelty

This study offers several unique contributions to the growing intersection of environmental informatics and digital technologies:

- It is one of the few reviews to explicitly focus on eutrophication monitoring, rather than general water quality or smart environment applications, ensuring relevance to ecological processes and nutrient dynamics.

- It introduces a clear taxonomy of system architectures—edge, cloud, and hybrid—and relates them to practical concerns such as latency, connectivity, and scalability in aquatic settings.
- The review identifies critical yet underreported challenges in existing literature, particularly regarding data privacy, microcontroller integration, and security implementation.
- Connecting system capabilities with eutrophication-specific needs (e.g., nutrient sensing, chlorophyll-a monitoring, early bloom detection), this review provides a forward-looking framework for advancing research and supporting real-world environmental management efforts.

2. Material and method

In this subsection, the study defines the methodology utilized to conduct a systematic review concentrating on the issues, challenges, and opportunities associated with the integration of IoT (Internet of Things) and Cloud Computing in eutrophication monitoring. The study is based on a review of literature issued over the last decade, from 2015 to 2025. To the best of the authors, no similar comprehensive review exists within this specific context and period, making this study a unique contribution to the field. The research methodology includes the careful selection of applicable peer-reviewed articles from key online databases, namely Scopus, Google Scholar, and Web of Science, ensuring a careful and reliable examination of the topic.

2.1. Eligibility criteria

A systematic review of all peer-reviewed and published research relevant to the applications, challenges, and benefits of IoT and Cloud Computing in eutrophication monitoring was conducted for analysis. The review exclusively considered scholarly works published in English between 2015 and 2025. To ensure the study's significance and quality, stringent inclusion criteria were applied, selecting only peer-reviewed papers that, without delay, explored the implementation, technological challenges, and opportunities of IoT and cloud-based systems in the context of water quality and eutrophication monitoring. Studies were required to expressly address practical applications, system benefits, and present a defined research structure or methodology related to these technologies. Research articles that did not align with these specific focus areas were excluded. The detailed inclusion and exclusion criteria for this review are presented in Table 2 (Alfahaid et al., 2025; Alquraish, 2025; Awais et al., 2025).

Table 2
Proposed Inclusion and Exclusion Criteria.

Criteria	Inclusion	Exclusion
Topic	Article papers focusing on A Review of IoT and Cloud Computing: Issues, Challenges, and Opportunities in Eutrophication Monitoring	Article papers not focusing on A Review of IoT and Cloud Computing: Issues, Challenges, and Opportunities in Eutrophication Monitoring.
Research Framework	The Articles must include a research framework or methodology for A Review of IoT and Cloud Computing: Issues, Challenges, and Opportunities in Eutrophication Monitoring.	Articles must exclude research framework or methodology for A Review of IoT and Cloud Computing: Issues, Challenges, and Opportunities in Eutrophication Monitoring.
Language	Must be written in English	Articles published in languages other than English
Period	Articles between 2015 to 2025	Articles outside 2015 and 2025

2.2. Information sources

To gather relevant studies for this review, a systematic search was conducted across several prominent online academic databases. Scopus, Google Scholar, and Web of Science were chosen for their comprehensive collections of peer-reviewed research related to IoT (Internet of Things) and Cloud Computing applications in environmental monitoring. By employing targeted keyword combinations, we ensured broad and thorough coverage of the most pertinent articles within the field of eutrophication monitoring. These databases provided access to high-quality academic journals and conference proceedings (Alfahaid et al., 2025; Alquraish, 2025; Awais et al., 2025; Chen et al., 2025; Ihimbazwe Ndanguza, 2025; Khanyi et al., 2024; La Cognata et al., 2025; Lanz et al., 2025; Le Roux et al., 2025; Miller et al., 2025; Mkhize et al., 2025; Mtjilibe et al., 2024; Pimenow et al., 2025; Rosário & Raimundo, 2025; Skosana et al., 2024; Teixeira et al., 2025; Thobejane & Thango, 2024). Collectively, they offered a solid foundation for this literature review, ensuring a well-rounded and comprehensive selection of relevant studies.

2.3. Search strategy

The literature for this review was gathered from reputable online academic databases, focusing on keywords that addressed both the technical and practical aspects of IoT (Internet of Things) and Cloud Computing in the context of eutrophication monitoring. To capture studies relevant to diverse monitoring systems and environmental management frameworks, the search incorporated terms such as "Eutrophication Monitoring", "Water Quality Monitoring", and "Algal Bloom Detection" alongside technological keywords. A systematic search was carried out across three primary repositories: Google Scholar, Scopus, and Web of Science. A carefully constructed search string was applied to identify the most relevant and up-to-date research. The keywords used were: ("IoT" OR "WSN") AND ("Cloud" OR "Edge" OR "Fog" OR "AWS" OR "Azure" OR "Google IoT" OR "ThingSpeak" OR "Blynk") AND ("Water Monitoring" OR "Algal Bloom" OR "Nutrient Pollution") AND ("Challenges" OR "Security" OR "Latency" OR

"Accuracy" OR "Energy") AND ("AI" OR "ML" OR "Blockchain" OR "5G" OR "Sustainable").To maintain relevance and reflect recent advancements, the search was limited to publications from 2015 to 2025. The initial search returned a substantial collection of potential sources: 5000 papers from Google Scholar, 263 from Scopus, and 5000 from Web of Science. Through a rigorous screening process, we then refined this collection by selecting only those studies that closely aligned with our research objectives. This systematic curation ensured that the review was built upon the most current, high-quality, and pertinent literature available in the field (Chen et al., 2025; Ihimbazwe Ndanguza, 2025; Khanyi et al., 2024; La Cognata et al., 2025; Lanz et al., 2025; Le Roux et al., 2025; Miller et al., 2025; Mkhize et al., 2025; Mtjilibe et al., 2024).

Table 3
Results Achieved from Literature Search.

No.	Online Repository	Number of results
1	Google Scholar	5000
2	Web of Science	481
3	Scopus	263
Total		5744

2.4. Selection process

Five researchers (MT, AS, AJT, MSS) independently reviewed the titles and abstracts of the first 60 records retrieved from the systematic search. Any discrepancies in article selection were addressed through group discussion until a consensus was achieved. Following this initial screening, the researchers proceeded in pairs to independently review the titles and abstracts of all remaining records. In instances where disagreements occurred, discussions were held to decide which studies should advance to full-text evaluation. If a resolution could not be reached within the pair, a third researcher was consulted to make the final determination. Subsequently, four researchers (MT, AS, AJT, MSS) independently assessed the full-text articles to verify whether they met the predetermined inclusion criteria for studies on IoT and Cloud Computing applications in eutrophication monitoring. As before, any differences in judgment were resolved through open discussion. Where necessary, the fifth researcher (BAT) provided the deciding input on whether to include or exclude specific articles. The full screening and selection process is illustrated in Fig. 1 (Alfahaid et al., 2025; Alquraish, 2025; Awais et al., 2025; Chen et al., 2025; Ihimbazwe Ndanguza, 2025; Khanyi et al., 2024; La Cognata et al., 2025; Lanz et al., 2025; Le Roux et al., 2025; Miller et al., 2025; Mkhize et al., 2025; Mtjilibe et al., 2024; Pimenow et al., 2025; Rosário & Raimundo, 2025; Skosana et al., 2024; Teixeira et al., 2025; Thobejane & Thango, 2024).

2.5. Data collection process

To ensure the accuracy and reliability of the data collected from the selected studies, we implemented a structured and rigorous approach designed to minimize errors and reduce potential bias. Two reviewers

independently extracted data from each included study, under the supervision of a third reviewer. Any discrepancies between their findings were discussed thoroughly until a consensus was reached. To maintain consistency across the research team, a standardized data extraction form. No automated software tools were used for data extraction; instead, all data entries were performed manually and independently double-checked to ensure precision and accuracy (Alfahaid et al., 2025; Alquraish, 2025; Awais et al., 2025; Chen et al., 2025; Ihimbazwe Ndanguza, 2025; Khanyi et al., 2024; La Cognata et al., 2025; Lanz et al., 2025; Le Roux et al., 2025; Miller et al., 2025; Mkhize et al., 2025; Mtjilibe et al., 2024; Pimenow et al., 2025; Rosário & Raimundo, 2025; Skosana et al., 2024; Teixeira et al., 2025; Thobejane & Thango, 2024).

In instances where study information was incomplete or unclear, a comprehensive review of all available supporting materials-including supplementary files, appendices, and related publications conducted to clarify the necessary data. If ambiguity persisted, the expertise of a fifth reviewer, a subject matter specialist in IoT and environmental monitoring technologies, was sought to validate the interpretation and uphold the credibility of the review process.

When multiple publications from the same study were identified, we applied strict selection criteria to prioritize the most recent and comprehensive reports published between 2015 and 2025. In cases where data discrepancies existed across these reports, we conducted a careful examination of the methodologies, findings, and reported results to reconcile differences and ensure data integrity.

Additionally, to maintain consistency in analysis and avoid potential misinterpretation due to language differences, only studies published in English were considered eligible for inclusion in this review. This comprehensive data extraction and validation process is summarized in Fig. 2 (Alfahaid et al., 2025; Alquraish, 2025; Awais et al., 2025; Chen et al., 2025; Ihimbazwe Ndanguza, 2025; Khanyi et al., 2024; La Cognata et al., 2025; Lanz et al., 2025; Le Roux et al., 2025; Miller et al., 2025; Mkhize et al., 2025; Mtjilibe et al., 2024; Pimenow et al., 2025; Rosário & Raimundo, 2025; Skosana et al., 2024; Teixeira et al., 2025; Thobejane & Thango, 2024).

2.6. Data items

This subsection provides a detailed look at the key findings from our systematic review, exploring how IoT and Cloud Computing technologies are being used to monitor eutrophication. We focused on several critical factors, including system performance, data accuracy, energy efficiency, scalability, response times, and security measures (Alfahaid et al., 2025; Alquraish, 2025; Awais et al., 2025; Chen et al., 2025; Ihimbazwe Ndanguza, 2025; Khanyi et al., 2024; La Cognata et al., 2025; Lanz et al., 2025; Le Roux et al., 2025; Miller et al., 2025; Mkhize et al., 2025; Mtjilibe et al., 2024; Pimenow et al., 2025; Rosário & Raimundo, 2025; Skosana et al., 2024; Teixeira et al., 2025; Thobejane & Thango, 2024).

Beyond these technical aspects, we also examined broader elements like study setups, monitoring locations, the technologies used, real-world deployment conditions, and common operational barriers. To give a complete picture, we considered cost-effectiveness, environmental factors, regulatory impacts,

and future possibilities, helping us understand how these systems perform in practice and what challenges they face.

2.6.1 Data Collection Method

This review is designed to provide a clear and well-rounded understanding of how IoT and Cloud Computing technologies are shaping the way we monitor eutrophication and overall water quality. We focused on identifying and defining key outcomes that highlight the technical, operational, and environmental impacts of these systems in aquatic monitoring. Our outcomes centred around several important areas. First, System Performance and Data Accuracy stood out as major outcomes. We looked at how these technologies enhanced the speed and reliability of data transmission, the sensitivity of sensors, and the accuracy of nutrient level readings. These aspects were essential in appraising how well IoT and cloud solutions enhance the quality of environmental data. Energy Efficiency and Scalability were also key focus areas. We evaluated how much energy the systems consumed during sensing and data transmission, and how easily the technologies could be expanded to cover multiple monitoring locations. These metrics helped us understand the practicality and long-term viability of large-scale IoT monitoring systems. We also assessed Latency and Data Processing Speed by analysing system responsiveness and how effectively cloud or edge computing reduced delays in data collection and analysis. Studies that demonstrated faster or near real-time processing were particularly valuable for this review. Security and Privacy were treated as critical concerns. We examined how data is protected, how vulnerabilities are managed, and what strategies are in place to ensure the user and environmental data remain secure. Priority was given to studies that addressed these issues with robust frameworks and solutions. Lastly, we explored Opportunities and Future Trends, paying close attention to how emerging technologies like AI, machine learning, blockchain, and 5G are being integrated into these systems. This helped us capture a forward-looking view of how IoT and cloud platforms can evolve to become more intelligent, reliable, and sustainable for environmental monitoring.

2.6.2 Definition of Collected Data Variables

In addition to the primary outcomes, we carefully considered several contextual variables to provide a well-rounded understanding of how IoT and Cloud Computing technologies are applied in eutrophication monitoring. These variables were essential for contextualizing findings and assessing the broader implications of technology adoption in environmental monitoring systems. Study characteristics such as geographical location, type of water bodies (e.g., lakes, rivers, reservoirs), and monitoring objectives were recorded to evaluate the applicability of findings across diverse ecological and operational settings. Intervention characteristics were also documented, detailing the types of IoT devices, sensors, cloud services, and data processing frameworks used, as well as their integration with existing monitoring infrastructures. Additionally, we considered economic factors, including installation costs, operational expenses, and reported cost-benefit outcomes, to assess the financial viability of deploying these technologies. External influences such as environmental policies, regulatory frameworks, and regional water management strategies were also reviewed to understand factors affecting implementation and

system effectiveness. As shown in Table 4, our systematic search across Google Scholar, Scopus, and Web of Science was tailored to capture the most relevant, high-quality studies (Alfahaid et al., 2025; Alquraish, 2025; Awais et al., 2025; Chen et al., 2025; Ihimbazwe Ndanguza, 2025; Khanyi et al., 2024; La Cognata et al., 2025; Lanz et al., 2025; Le Roux et al., 2025; Miller et al., 2025; Mkhize et al., 2025; Mtjilibe et al., 2024; Pimenow et al., 2025; Rosário & Raimundo, 2025; Skosana et al., 2024; Teixeira et al., 2025; Thobejane & Thango, 2024). This methodical and structured approach ensures that our review offers a reliable and comprehensive assessment of the challenges, limitations, and opportunities associated with IoT and Cloud Computing in eutrophication.

Table 4
Data Variables Collected.

Field	Description
Study characteristics	Geographic region, type of water body (e.g., lake, river, reservoir), and the primary monitoring goals to contextualize the scope of each study.
Intervention characteristics	Types of IoT devices, sensors, cloud/edge platforms, and data processing frameworks used, including how they integrate with existing systems.
Economic factors	Installation costs, operational expenses, and reported cost-benefit outcomes reflecting financial feasibility and sustainability.
External influences	Environmental regulations, regional policies, and water management strategies that impact implementation and system effectiveness.

2.7 Study risk of bias assessment

To evaluate the robustness of the synthesized results in this review, we conducted a sensitivity analysis by selectively modifying key inclusion variables and re-analyzing the dataset. This involved examining how the overall trends and conclusions were affected when specific categories of studies were excluded or emphasized, such as:

- Studies lacking clearly defined latency or real-time response metrics
- Papers without stated security or privacy mechanisms
- Studies using proprietary cloud platforms versus open-source or custom-built alternatives
- Research that did not integrate microcontrollers or edge devices
- Studies from underrepresented regions or with unclear methodological frameworks

This analysis aimed to determine whether the core conclusions, such as edge-cloud hybrid systems' effectiveness, latency's impact on real-time monitoring, and recurring scalability challenges, remained valid across different study subsets. Results remained consistent in most cases, suggesting that the findings were not unduly influenced by outliers or methodological bias in any particular group of studies. This methodological filtering helped confirm the synthesized results' reliability and stability. It highlighted the importance of transparency and completeness in future reporting of IoT-based eutrophication monitoring systems.

2.8 Effect measures

Effect measures as shown in Fig. 3 were used to present the results obtained from the analysis of the systematic review data related to the application of IoT and cloud computing technologies in eutrophication monitoring. The effect measurement process begins by defining precise research questions that direct the systematic data collection. This is followed by applying appropriate criteria and metrics aligned with the intended search outcomes, such as the cloud platform utilized, cloud-based data processing architecture, and security mechanisms. The findings are reported through tables, graphs, and charts to ensure consistency and clarity in the presentation of results. Subsequently, a meta-analysis is conducted to assess the variability and variety associated with the studied data.

2.9 Synthesis Analysis

2.9.1 Study eligibility for synthesis

The criteria used for analysing the systematic literature review were drawn directly from all the included eligible research papers. This information included the year of publication (ranging from 2015–2025), the country of the authors, the research source or repository, and the type of research conducted. Further analysis was based on the cloud platform utilized, the method of data upload, the data storage mechanisms employed, and performance metrics such as latency and real-time response in edge versus cloud processing architectures. Additional parameters considered were the security mechanisms implemented, privacy concerns identified, their corresponding mitigation strategies, integration with microcontrollers, as well as scalability challenges, and opportunities for future expansion. All selected papers adhered to the inclusion and exclusion criteria outlined in Section 2.1, Table 2, and met the risk of bias assessment requirements described in Section 2.7 Study risk of bias assessment.

2.9.2 Data preparation method

To organize the data for analysis, we used a structured Excel spreadsheet to systematically categorize all key elements extracted from the reviewed research papers. This approach also helped in identifying any missing information that could impact the accuracy of the review. To highlight and mark out the missing information within the research papers that met the inclusion criteria, we used "Not specified" as an indicator.

2.9.3 Tabulation and display method

To effectively summarize and synthesize the findings from the reviewed studies, we used multiple methods to tabulate and visually present the results. Initially, all relevant data extracted from each eligible study were organized in a structured Excel spreadsheet. The Excel Data Analysis Toolbox was then used to generate visual representations of the data, including bar graphs, pie charts, and line graphs. These included Security Mechanisms comparison, Edge vs Cloud Processing architecture distribution, and Scalability Challenges vs Opportunities for future expansion. These visuals helped

compare the prevalence and usage of various technologies, identify trends across years or regions, assess the distribution of security and privacy mechanisms, and highlight gaps in current research.

2.9.4 Synthesis Methods

To synthesize the results of the reviewed studies, a categorical synthesis approach was applied using Microsoft Excel. Studies were systematically classified based on predefined criteria, including publication year, country of origin, research type, cloud platform utilized, data upload methods, data storage mechanisms, edge vs. cloud processing architecture, latency and real-time performance, security mechanisms, privacy concerns and their mitigation, integration with microcontrollers, and scalability considerations. This method facilitated the identification of recurring themes, patterns, and gaps within the literature. Due to the heterogeneous nature of the studies, covering diverse hardware setups, software platforms, and deployment contexts, a narrative synthesis was employed to describe and interpret the relationships between different study variables. Quantitative trends such as the frequency of specific technologies or architectures were highlighted using tabulated summaries and visualizations. Descriptive statistics were used to support comparisons across studies. Missing or unspecified data points were noted accordingly but excluded from comparative synthesis to maintain clarity and consistency. A qualitative and descriptive approach ensured a meaningful synthesis of diverse methodologies and outcomes. No statistical software was used due to the absence of directly comparable quantitative effect measures across the eligible studies.

2.9.5 Exploration of heterogeneity

To explore the possible causes of heterogeneity among the reviewed studies, subgroup analysis was conducted based on key study attributes (Alfahaid et al., 2025; Alquraish, 2025; Awais et al., 2025; Chen et al., 2025; Ihimbazwe Ndanguza, 2025; Khanyi et al., 2024; La Cognata et al., 2025; Lanz et al., 2025; Le Roux et al., 2025; Miller et al., 2025; Mkhize et al., 2025; Mtjilibe et al., 2024; Pimenow et al., 2025; Rosário & Raimundo, 2025; Skosana et al., 2024; Teixeira et al., 2025; Thobejane & Thango, 2024). These included the type of cloud platform used, data upload methods, data storage mechanisms, processing architectures, and latency metrics. Additional factors such as the country of the study, research type, integration with microcontrollers, and the presence or absence of security and privacy mechanisms were also analyzed. These subgroup analyses allowed for the identification of variation in implementation outcomes and helped to explain inconsistencies in real-time performance, scalability, and system responsiveness across different environmental and technical contexts. For example, studies using edge processing often reported lower latency compared to cloud-only solutions, particularly in remote or bandwidth-limited regions. Similarly, differences in privacy handling were observed between studies from regions with strict data protection regulations versus those without. By examining these variations, the review was able to uncover key contextual factors—such as technological maturity, regulatory environment, and infrastructure availability—that contribute to the diversity in outcomes. This approach also helped identify gaps in standardization and areas where further research or unified frameworks are needed. A qualitative interpretation of subgroup patterns was used to support the synthesis and guide future research directions.

2.9.6 Sensitivity analysis

To assess the robustness of the synthesized results, a sensitivity analysis was performed by selectively excluding or emphasizing studies based on specific variables such as latency and real-time response metrics, security mechanisms, cloud platform type, microcontroller integration, and scalability. This helped identify whether core findings, such as the advantages of hybrid processing or security mechanisms(e.g, authentication), remained consistent across different studies. The results confirmed that the main conclusions were stable and not heavily influenced by outliers or incomplete data.

2.10 Reporting bias assessment

To ensure the credibility and comprehensiveness of this systematic review, a risk of bias assessment was conducted to evaluate the potential for missing results due to selective reporting or publication practices. The selection of literature was limited to English-language publications, which may have excluded relevant studies published in other languages. Only openly accessible databases and repositories such as Web of Science, Scopus, and Google Scholar were used, potentially introducing bias toward more visible or widely cited studies. Moreover, a predefined set of attributes—such as cloud platform, data upload method, latency metrics, and privacy concerns—guided the data extraction process (Alfahaid et al., 2025; Alquraish, 2025; Awais et al., 2025; Chen et al., 2025; Ihimbazwe Ndanguza, 2025; Khanyi et al., 2024; La Cognata et al., 2025; Lanz et al., 2025; Le Roux et al., 2025; Miller et al., 2025; Mkhize et al., 2025; Mtjilibe et al., 2024; Pimenow et al., 2025; Rosário & Raimundo, 2025; Skosana et al., 2024; Teixeira et al., 2025; Thobejane & Thango, 2024). This may have resulted in the omission of studies that did not explicitly report on these parameters, even if their contributions were otherwise significant. Manual data analysis and tabulation in Excel were employed to reduce the likelihood of overlooking critical information, though the possibility of unreported outcomes remains.

3. Results

3.1. Study selection

For this systematic review, we followed a careful and structured process to gather and select studies that were relevant and of good quality. We started by searching three well-known online databases — Web of Science, Google Scholar, and Scopus. Altogether, this search produced 5,744 records, with 481 from Web of Science, 5,000 from Google Scholar, and 263 from Scopus. Next, we screened the titles and abstracts of all these records to check whether they matched our review's focus. After this first round, 5,595 studies were excluded because they didn't meet the criteria, leaving us with 148 studies we wanted to retrieve in full. Unfortunately, we couldn't get hold of 67 of these studies, so we moved forward with the 81 studies we managed to access. We then carefully reviewed the full text of these 81 studies to decide if they were suitable for our review. In the end, all 81 studies met our eligibility criteria and were included in the final review. The entire selection process is shown in Fig. 5, which breaks down each step from identifying studies to final inclusion.

Additionally, the distribution of articles obtained from each database is illustrated in Fig. 6, highlighting the contributions of the three databases to the final pool of studies. These visualizations enhance the transparency of the selection process and facilitate the replication of this methodology by other researchers.

The distribution of published research papers over the past decade demonstrates a progressive increase in scholarly output, with noticeable growth beginning in 2018 and peaking in 2023. As illustrated in Fig. 6, the number of studies remained modest between 2015 and 2017, followed by a marked upward trend from 2018 onward. The highest volume of publications occurred in 2023, indicating heightened academic and industrial interest during this period. This trend suggests a growing recognition of the field's relevance and possibly reflects broader advancements in enabling technologies and global research priorities.

A country-wise analysis of the reviewed articles reveals significant regional disparities in research contributions. As depicted in Fig. 7, the majority of the publications originate from Asia-Pacific countries, with India accounting for the largest share (33.33%), followed by China (13.58%). Other contributors include Malaysia, Vietnam, and Indonesia, collectively reinforcing the dominance of the Asia-Pacific region. Europe, North America, and the Middle East also present modest representation, with countries like Germany, the UK, and the USA contributing between 2–9% individually. Conversely, regions such as Africa and South America show minimal activity in this domain. This geographical distribution highlights the central role of Asian countries in driving research forward, possibly reflecting both regional technological priorities and developmental needs.

The sources of reviewed literature span multiple academic databases, reflecting a diversity of indexing platforms. As shown in Fig. 8, Google Scholar accounts for the largest proportion of articles at 57%, followed by Scopus with 29% and Web of Science at 14%. This distribution suggests a tendency among researchers to publish in open-access or broadly indexed repositories such as Google Scholar, potentially due to its inclusive coverage and accessibility. In contrast, the lower representation from Web of Science may reflect the more selective nature of that index.

An examination of cloud platform usage across the reviewed studies reveals a broad spectrum of deployment preferences, from custom solutions to well-established commercial offerings. As presented in Fig. 9, ThingSpeak emerges as the most frequently adopted platform, accounting for 30.86% of the cases, followed by Custom Cloud (9.88%) and Generic Cloud solutions (8.64%). A substantial portion of studies (23.46%) did not specify the cloud service utilized, indicating either omitted information or platform-agnostic architectures.

Platforms were further categorized into six functional classes: Infrastructure (e.g., IaaS), Commercial Cloud Platforms (e.g., Azure, IBM Watson IoT), Custom/Generic Platforms, IoT-Specific Platforms (e.g., FIWARE, ThingWorx), Database/Storage solutions (e.g., MySQL), and Unspecified. This distribution underscores a strong inclination toward lightweight, IoT-dedicated services and tailor-made

deployments, suggesting the importance of flexibility and domain-specific functionality in IoT-driven systems.

The reviewed studies exhibit a diverse range of communication protocols and data transmission methods, reflecting the heterogeneity of IoT architectures. As shown in Fig. 10, HTTP is the most commonly used protocol, accounting for 33.33% of implementations, followed by MQTT (12.35%) and REST-based APIs (9.87%). Hybrid combinations such as MQTT with HTTP (6.17%) and HTTP with API (4.94%) indicate a trend toward protocol layering to meet specific system requirements for reliability, speed, or interoperability. Notably, 29.63% of the studies did not specify the communication method employed, which may point to documentation gaps or the use of off-the-shelf systems where protocols are abstracted.

The protocols are further categorized into Web Protocols, IoT-specific Protocols (e.g., MQTT), RESTful APIs, and various integrated stacks (e.g., IoT + Web + REST). This diversity underscores the flexibility of communication design in IoT deployments and highlights the influence of application demands on protocol selection.

The storage mechanisms employed across the reviewed studies reflect a wide variety of implementation strategies and system requirements. As depicted in Fig. 11, the most common storage solution falls under the “Other” category (35.80%), indicating the use of unconventional, unclassified, or domain-specific methods. Real-time databases account for 20.99%, suggesting their importance in applications requiring rapid data acquisition and access. Traditional SQL-based relational databases follow with 13.58%, while JSON—a lightweight, NoSQL-compatible format—is used in 11.11% of cases.

A notable 14.81% of studies do not specify the storage mechanism utilized, and a smaller share (3.70%) mention the use of multiple integrated storage methods. The mechanisms are categorized broadly into Real-time Databases, Relational Databases, Lightweight/NoSQL formats, Hybrid/Combined approaches, and Unclassified/Miscellaneous solutions. This spread underscores the diversity of storage requirements in IoT systems, influenced by data volume, latency needs, and system architecture.

Platform deployment preferences among the reviewed systems indicate a clear leaning toward hybrid configurations. As illustrated in Fig. 12, hybrid platforms—those combining both edge and cloud capabilities—are the most dominant, representing 55.56% of the surveyed implementations. This preference reflects the increasing importance of balancing real-time responsiveness at the edge with the scalability and processing power of the cloud.

Edge-only platforms account for 28.40%, suggesting significant traction for low-latency, localized processing models, particularly in time-sensitive or connectivity-constrained environments. Cloud-only deployments comprise 13.58%, likely favoring centralized management and simplified infrastructure. A small fraction (2.47%) of studies did not specify the underlying platform model. This distribution highlights the architectural shifts in modern IoT systems, favoring flexible, distributed approaches over singular deployment models.

The analysis of security mechanisms employed across the reviewed studies reveals a concerning inconsistency in how security is addressed. As shown in Fig. 13, nearly half of the reviewed works (49.38%) did not specify any security mechanism, underscoring a significant gap in documentation or implementation. Among those that did report security measures, encryption was the most common (24.69%), followed by authentication techniques (16.05%). A smaller portion (9.88%) included only limited or implied security features, such as general system hardening or access control without explicit technical detail.

This distribution indicates that while some systems are built with a security-conscious design, a substantial number either overlook or underreport the protection mechanisms in place. Given the increasing exposure of IoT systems to cyber threats, this represents an important area for future research and standardization.

An analysis of microcontroller adoption reveals a strong preference for the Arduino ecosystem, which accounts for 34.57% of the reviewed systems. As shown in Fig. 14, a significant portion of studies (28.40%) did not specify the microcontroller used, reflecting either the use of high-level abstractions or omissions in reporting. The ESP family of devices—especially the ESP32 (17.28%) and ESP8266 (4.94%)—also shows considerable usage, likely due to their cost-effectiveness and built-in wireless capabilities.

Raspberry Pi appears in 9.88% of studies, supporting its role as a versatile yet more computationally intensive platform. Mixed deployments, such as combined use of Arduino and NodeMCU, account for a smaller share (2.47%), indicating occasional hybrid architectures. The categorization in the figure further groups microcontrollers into families, highlighting that ESP-based boards (including NodeMCU variants) collectively represent a significant share under the "ESP Family" category

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Figures

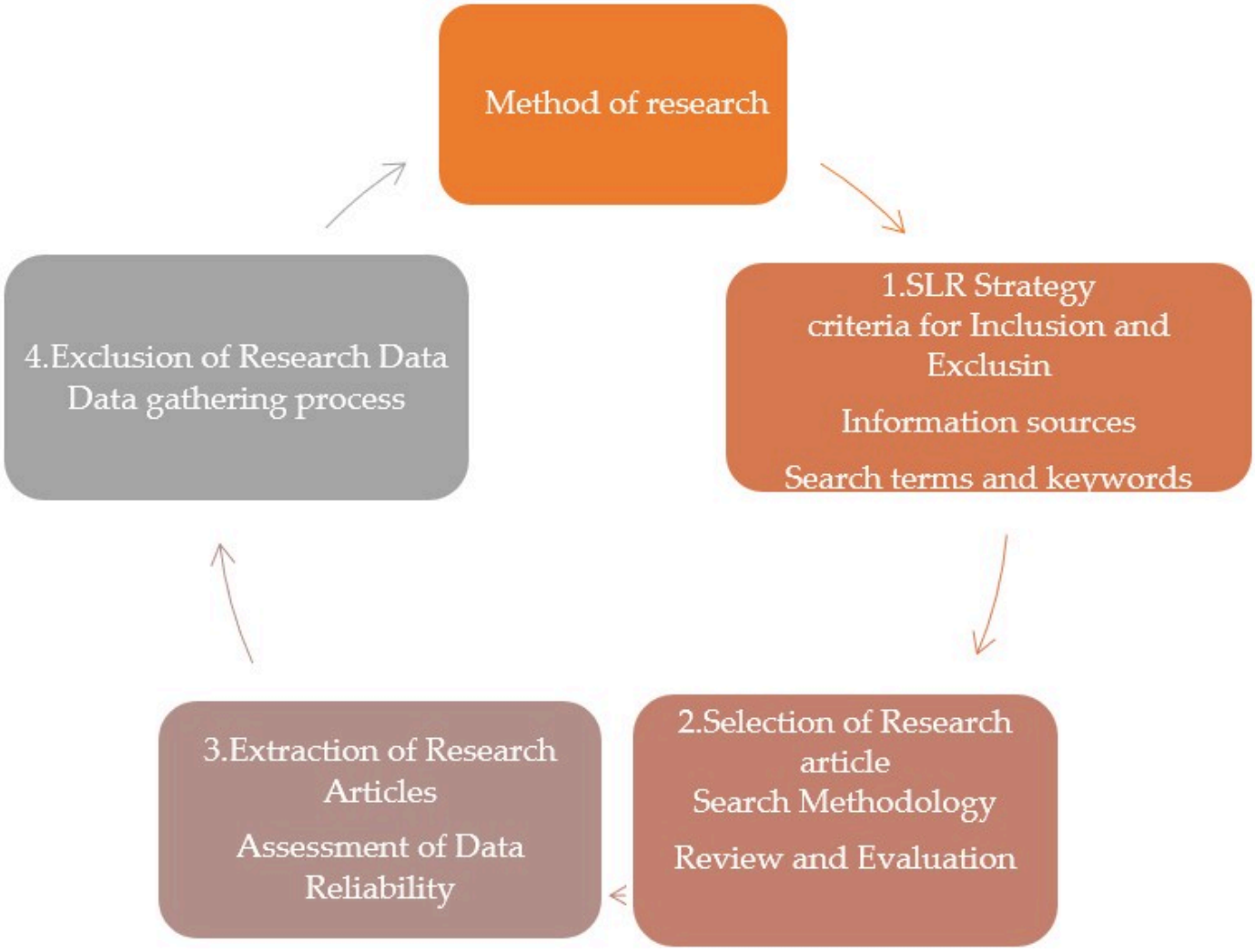

Figure 1

Procedures and stages of the review.

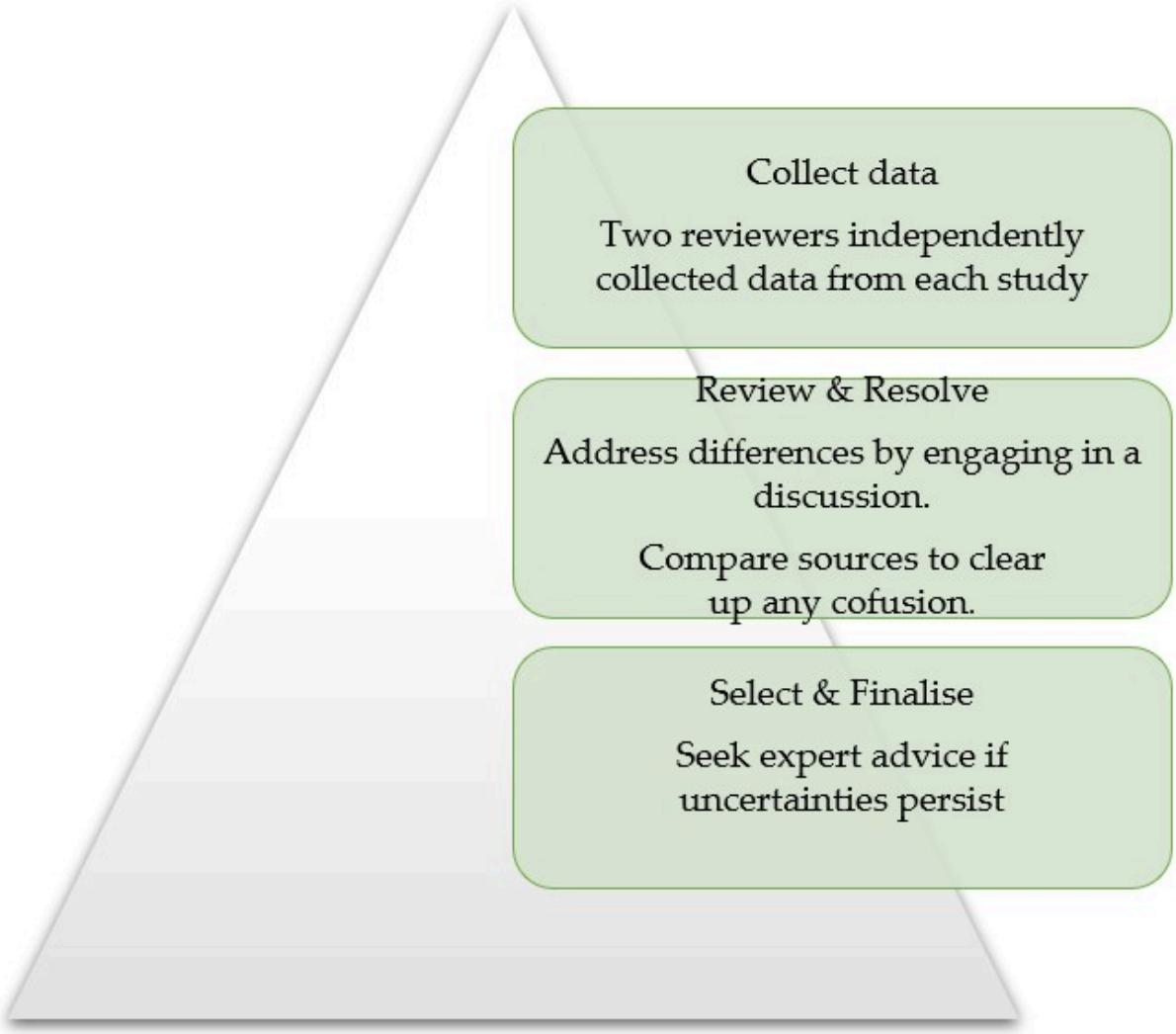

Figure 2

Flow of data selection and extraction

Figure 3

Effect measures process.

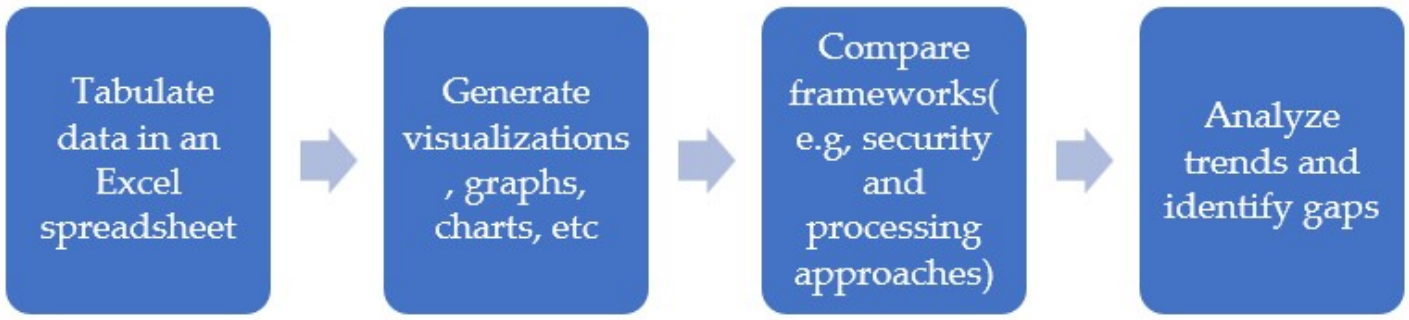

Figure 4

Tabulation and display method.

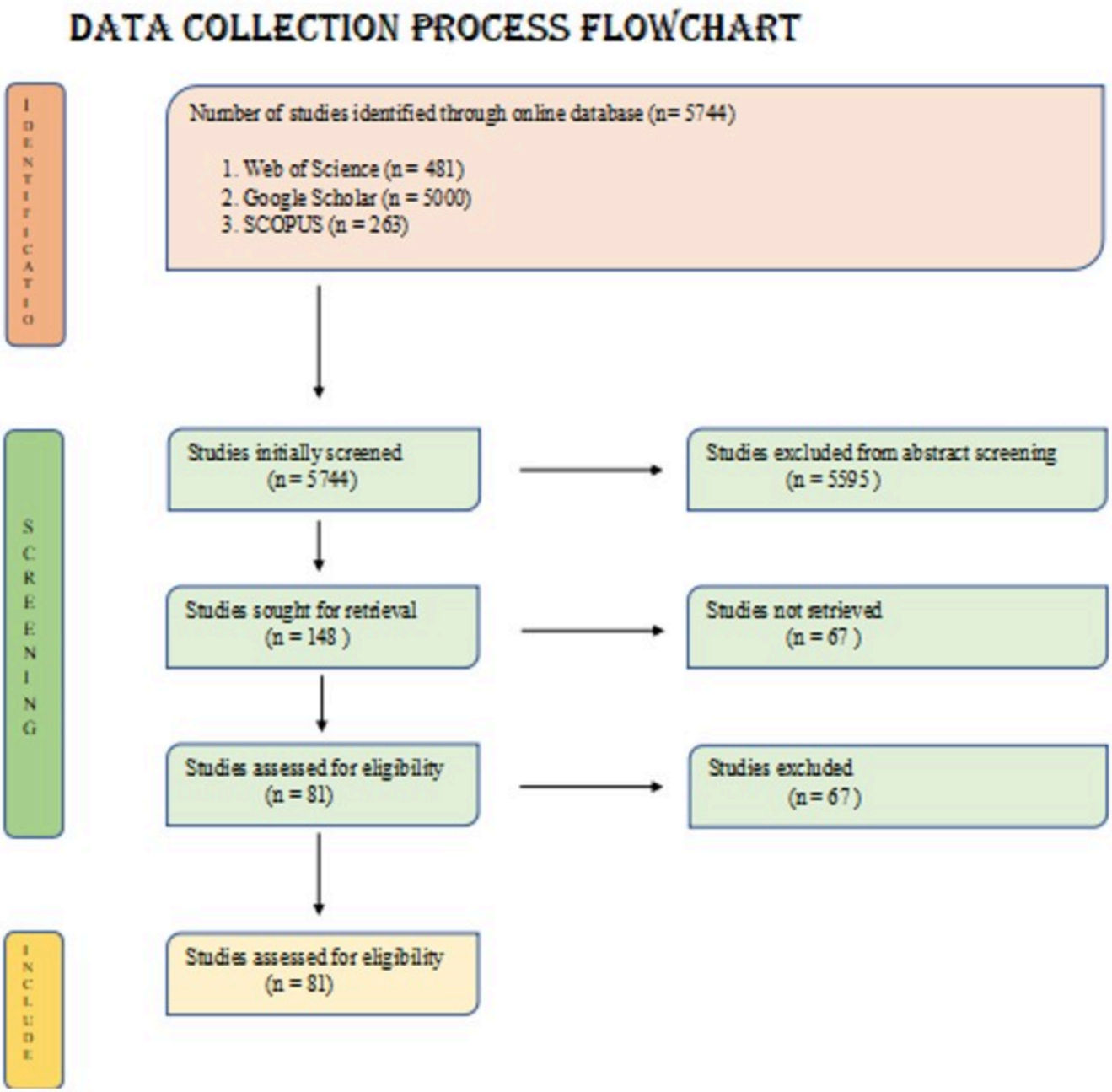

Figure 5

Proposed PRISMA Flowchart.

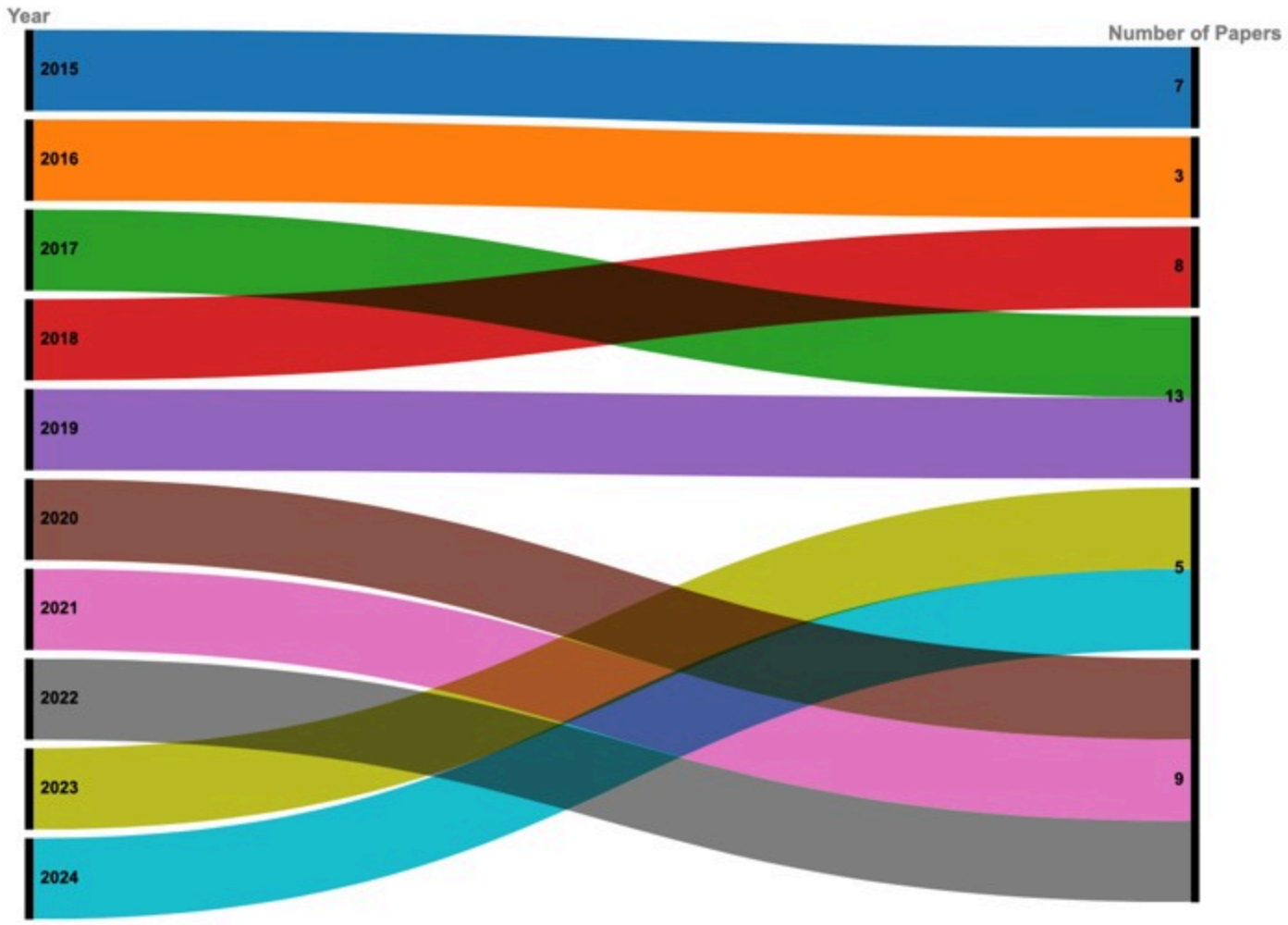

Figure 6

Distribution of published research papers.

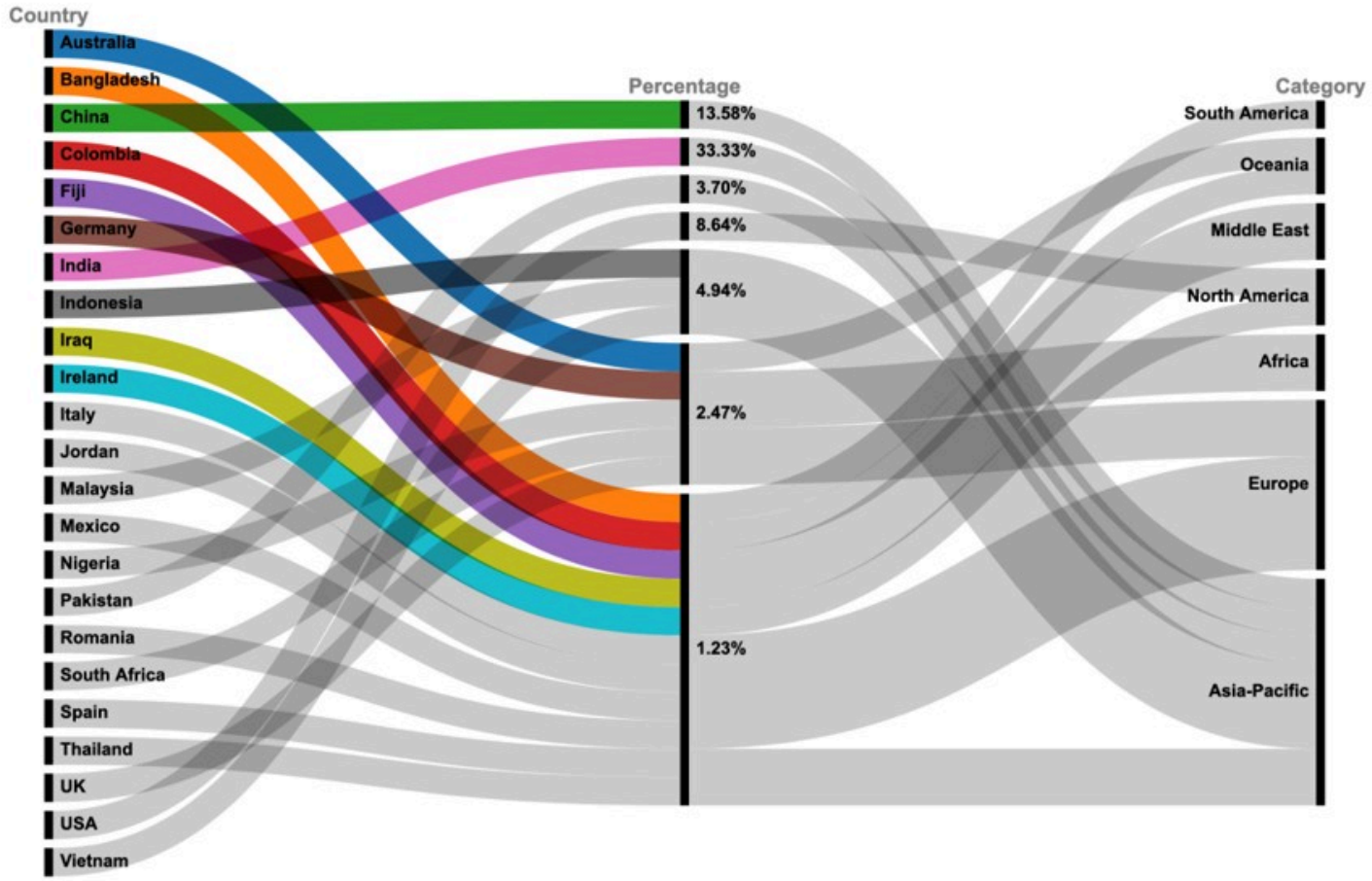

Figure 7

Country-wise distribution of reviewed articles.

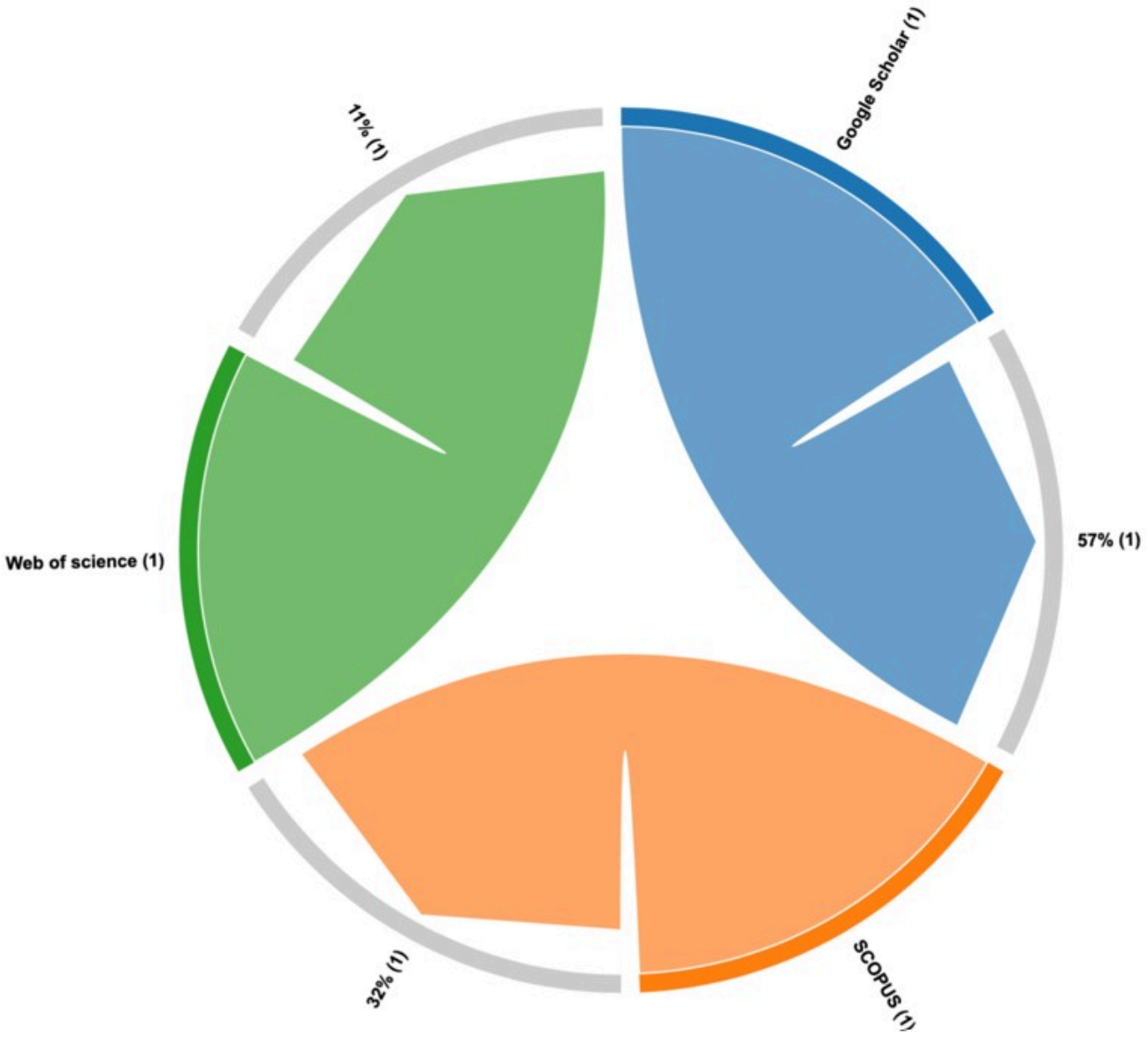

Figure 8

Distribution by Country.

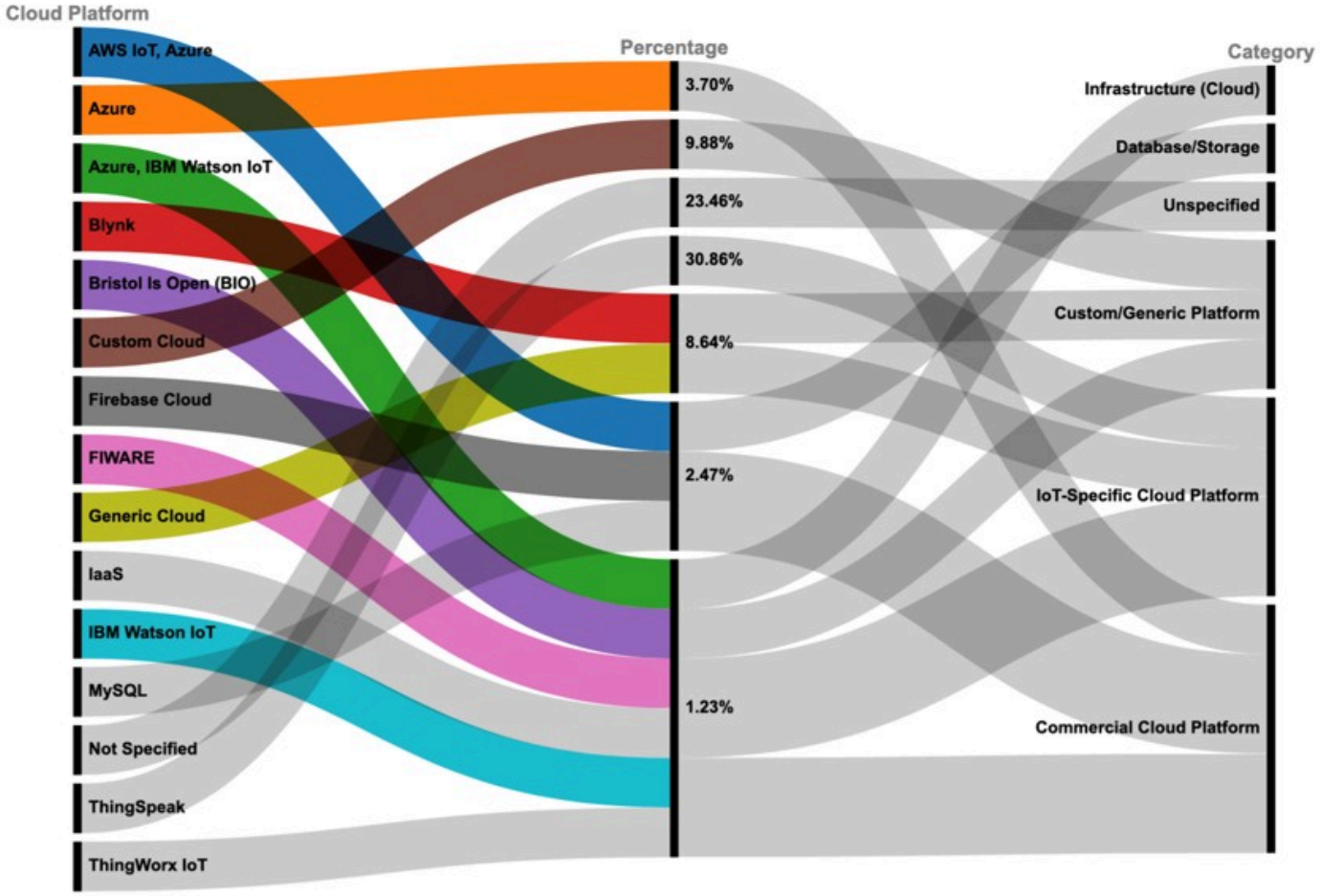

Figure 9

Distribution of cloud platforms.

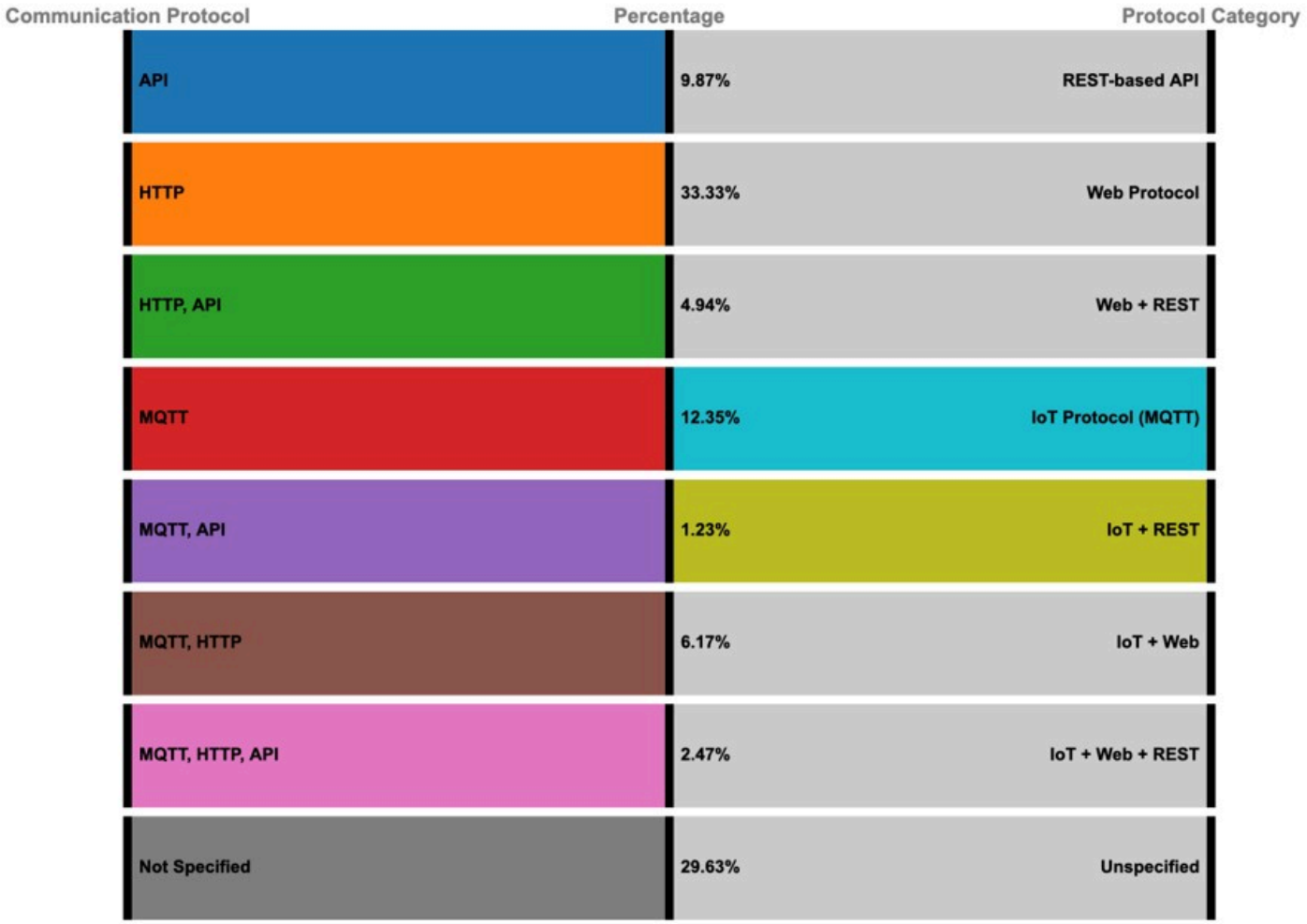

Figure 10

Communication protocols/data upload methods.

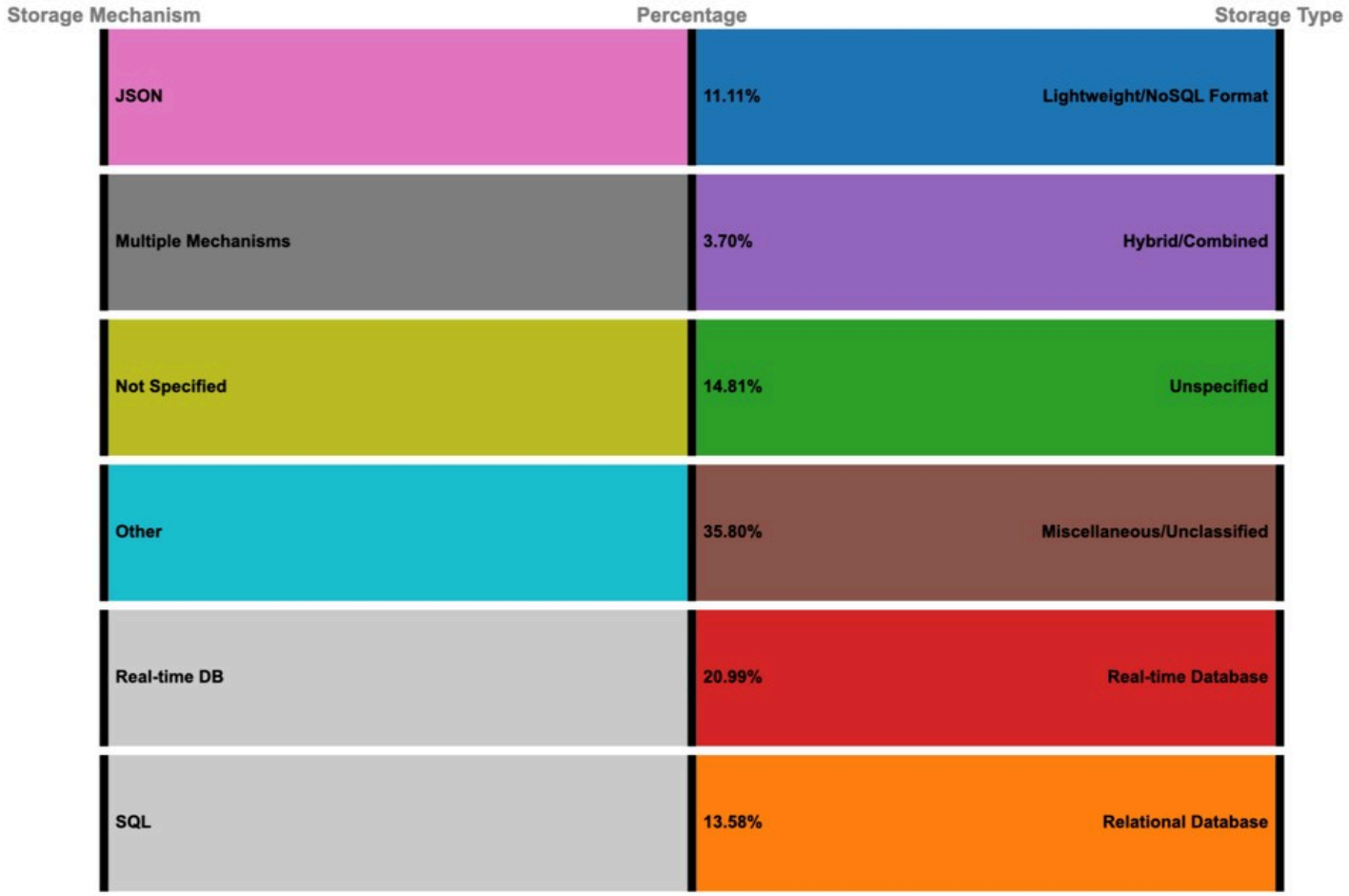

Figure 11

data storage mechanisms

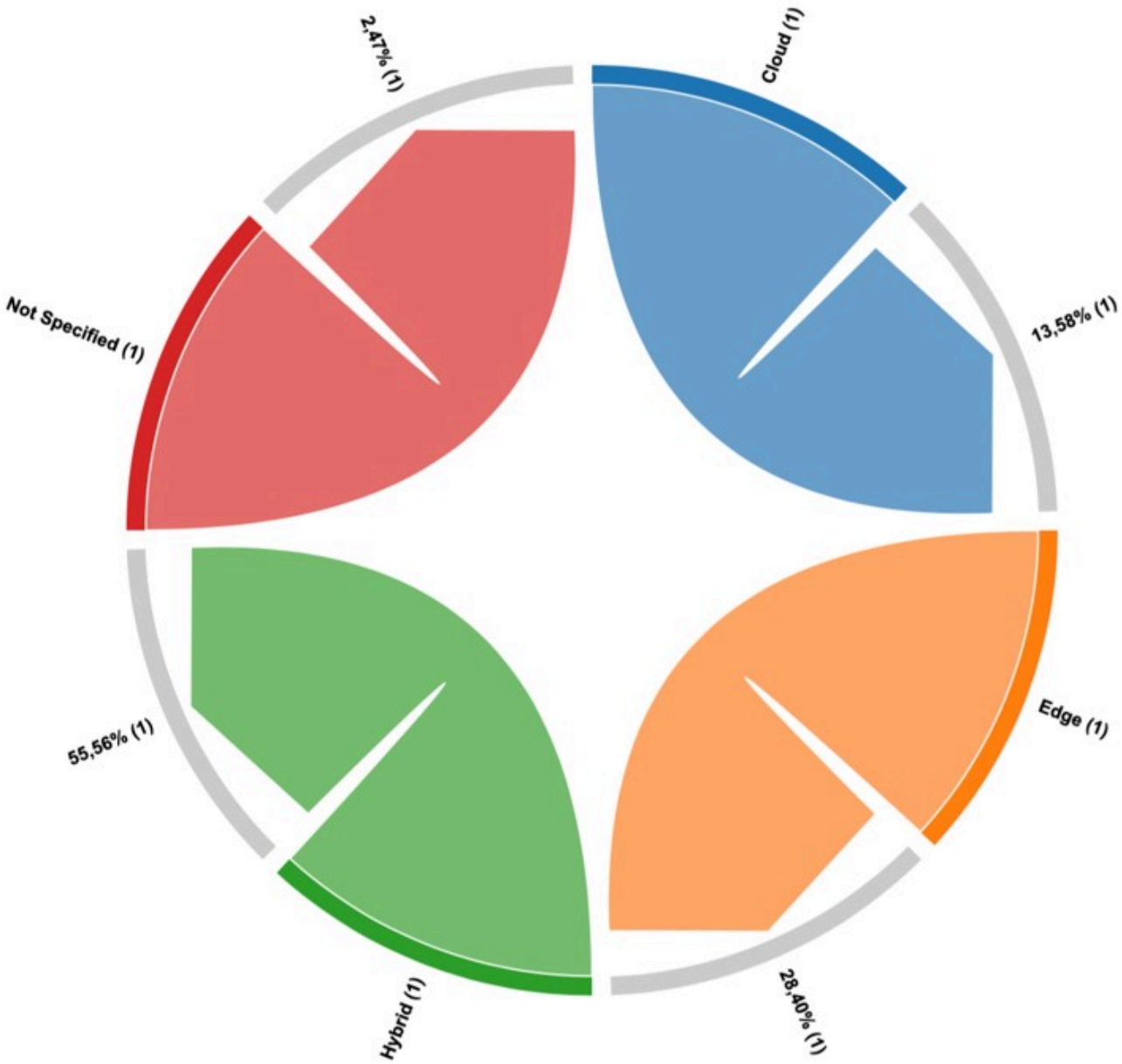

Figure 12

Platform Distribution.

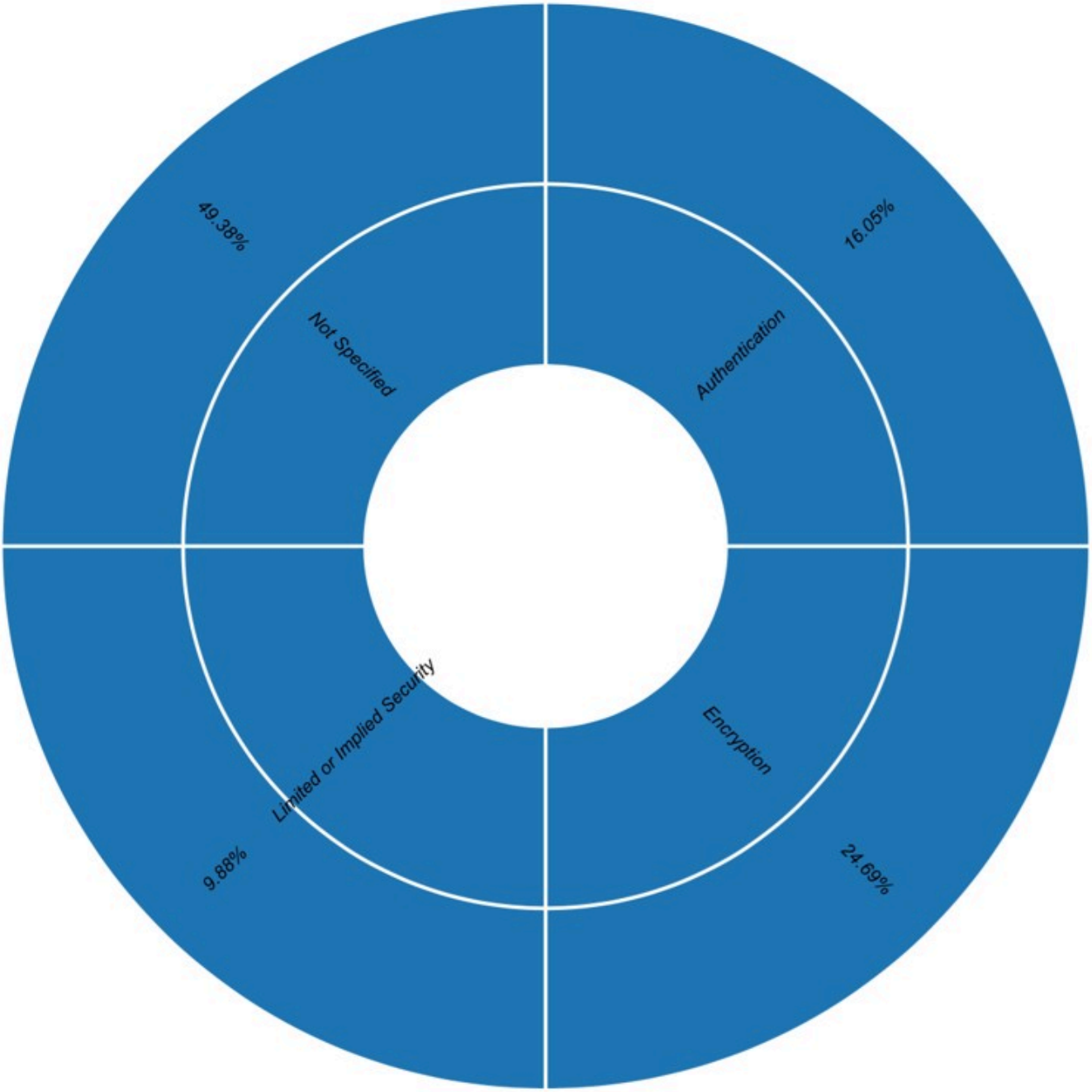

Figure 13

Security Mechanisms.

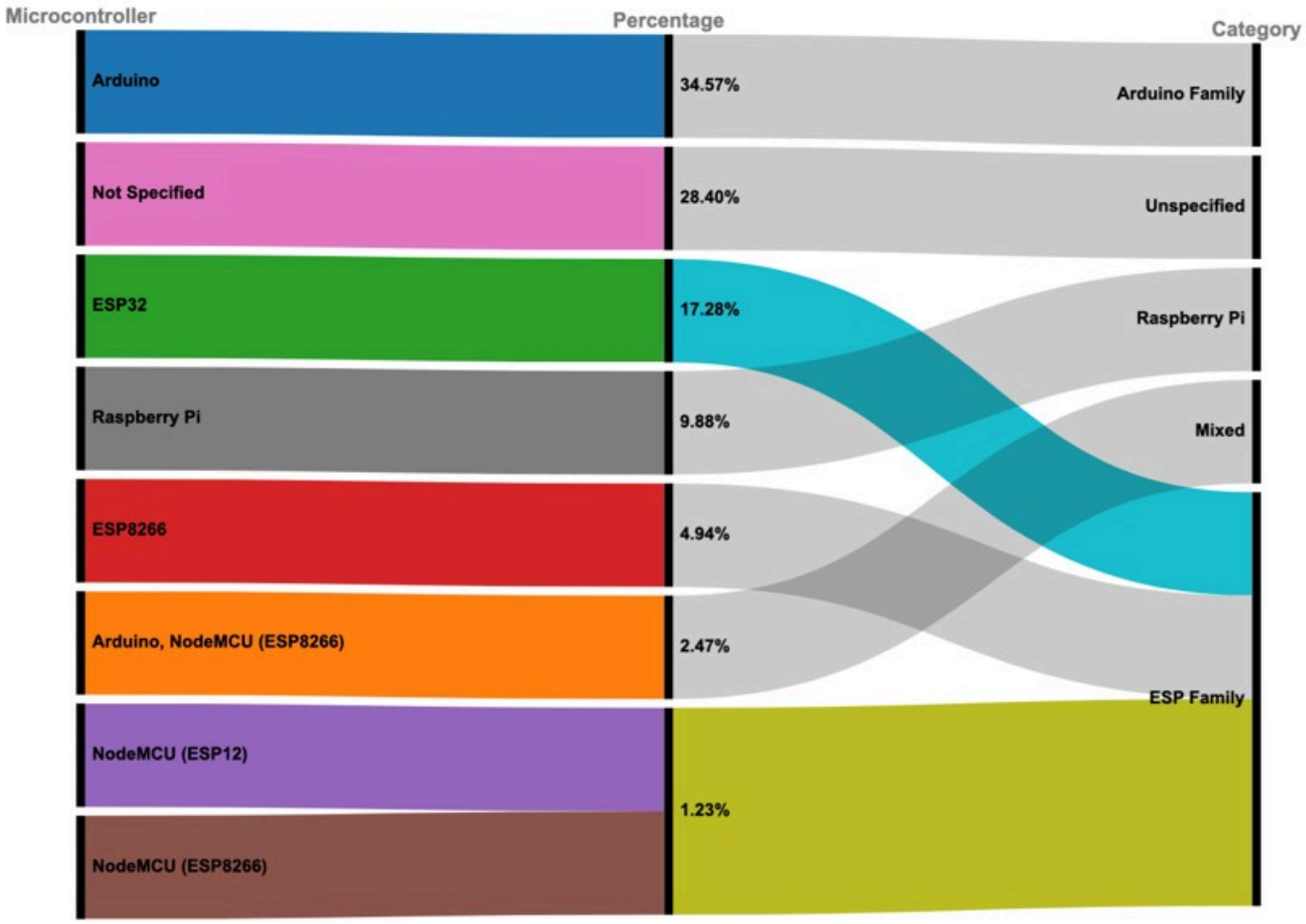

Figure 14

Microcontroller usage.